# Soroush Mahdi

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## *Education*

**Master of Science in Artificial Intelligence** — *Tehran, Iran*
*AmirKabir University of Technology (Tehran Polytechnic)* — *Sep 2020 – Sep 2023*
*Ranked **$7_{th}$** among 50 M.Sc. students; cumulative **GPA 17.61/20** (WES-equivalent: A / 4.0)*
*Thesis: Improving the robustness of deep neural network image classifiers against adversarial examples*

**Bachelor of Computer Engineering** — *Hamedan, Iran*
*Bu-Ali Sina University* — *Sep 2016 – Sep 2020*

## *Research Interests*

**World Models & Predictive Video Models**
**Memory & Long-Horizon State in Streaming Models**
**Embodied AI & Model-Based RL**
**Adaptation under Distribution Shift**

## *Publications*

1. **Mahdi, S.**, Ayar, F., Javanmardi, E., Tsukada, M., Javanmardi, M., "Evict3R: Training-Free Token Eviction for Memory-Bounded Streaming Visual Geometry Transformers". arXiv:2509.17650, 2025. *(Lead author; open-source.)*
2. Dastmalchi, H., **Mahdi, S.**, Alimoradijazi, M., Tamjidi, M., Saberi, M., Golzan, M., An, A., Cheraghian, A., "MODE-TTA: Multi-Mode Density Modeling for Test-Time Adaptation of 3D Vision–Language Foundation Models." Under review at ICLR 2027.
3. **Mahdi, S.**, Amirmazlaghani, M., Saravani, S., Dehghanian, Z., "MemLoss: Enhancing Adversarial Training with Recycling Adversarial Examples". arXiv:2510.09105, 2025. *(Lead author; from M.Sc. thesis.)*

## *Research Experience*

**Research Assistant** — *Tehran, Iran*
*Autonomous & Intelligent Systems Lab, AmirKabir University of Technology* — *Jun 2025 – Present*
*Under the supervision of Mahdi Javanmardi (with the University of Tokyo)*

- Developed **Evict3R** (lead author), on **what a streaming 3D model needs to remember**. Streaming visual geometry transformers (StreamVGGT) keep every past frame in a KV cache, so memory grows without bound over a video.
- Measured how attention concentrates layer by layer, and used that profile to allocate a **per-layer memory budget** and evict the least-attended tokens at inference, **with no retraining**.
- Peak GPU memory from **18.63 GB to 9.39 GB** at a 0.003 drop in reconstruction accuracy and completeness, enabling **2×–10× longer sequences** on one GPU; evaluated on 3D reconstruction, video depth and camera pose estimation (7-Scenes, NRGBD, Sintel, KITTI, TUM). Released as architecture-agnostic open-source code.

**Research Assistant** — *Remote*
*University of Sydney* — *Jan 2025 – Present*
*Under the supervision of Ali Cheraghian*

- Co-developed **MODE-TTA**: adapts a frozen 3D vision–language foundation model to a data stream that drifts from its training distribution, **without labels and one batch at a time**. Each class is a Gaussian mixture in feature space, updated online by hierarchical EM; a residual module realigns the frozen text embeddings as the distribution moves.
- Designed and implemented the full pipeline and ran all experiments; consistently outperformed prior test-time adaptation methods while keeping about 97% of the base model's throughput.

- Currently leading the problem framing and experiment design of a study of **hallucination in large vision–language models**.

**M.Sc. Thesis Research Assistant** *Tehran, Iran*
*Statistical Data Analysis Laboratory, AmirKabir University of Technology* *Jun 2021 – Sep 2023*
*Under the supervision of Maryam Amirmazlaghani*

- Proposed **MemLoss**: adversarial training that stores adversarial examples from earlier epochs and reuses them, improving both clean and robust accuracy with TRADES and HAT on CIFAR-10, CIFAR-100 and SVHN.

## *Industry Experience*

- **Data Scientist, Iran Credit Scoring** (Jul 2024 – Jun 2025): built and deployed machine-learning credit-scoring models in production under regulatory and data-drift constraints.
- **Computer Vision Researcher, HARA.ai** (2022): face anti-spoofing and liveness detection with deep learning.

## *Honors & Awards*

- **1st** Place, The Iran Annual AI Award, 2024 — deep-learning brain-abnormality diagnosis from MRI volumes.
- **2nd** Place, The National fNIRS Data Analysis Competition, National Brain Mapping Laboratory, 2022.
- Ranked **61st** (AI) and **36th** (Algorithms) among 15,000 (**top 0.4%**) in the Iranian M.Sc. Entrance Exam, 2020; **22nd** in the National Scientific Olympiad in Computer Engineering, 2020; West Asia Regional **ACM-ICPC**, 2017 & 2019.

## *Teaching Experience*

- Teaching assistant for 2 graduate courses at AUT (Statistical Machine Learning, Machine Vision) and 5 undergraduate courses at Bu-Ali Sina University, two of them including lectures (2017–2023).

## *Skills*

- **Methods:** transformer internals (attention, KV caching, memory-bounded inference); streaming 3D reconstruction (VGGT family, CUT3R, DUSt3R-style pointmap models); vision–language and 3D foundation models; test-time adaptation; mixture models and EM; GPU-scale experimentation
- **Mathematics:** probability and stochastic processes (19.28/20), engineering probability & statistics (19.5/20), statistical machine learning, optimisation
- **Programming: Python** (primary), C++, R | **Frameworks: PyTorch** (primary), JAX, TensorFlow, NumPy, SciPy, scikit-learn, OpenCV, Hugging Face; Git, Linux, LaTeX
- **Languages:** English (TOEFL iBT 103/120 — C1), Persian (native), German (currently taking A1 classes)

## *References*

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| Mahdi Javanmardi (Assistant Professor, AUT — research supervisor, Evict3R) | mjavan@aut.ac.ir | Homepage |
| Ali Cheraghian (Adjunct Lecturer, Univ. of Sydney — research supervisor, MODE-TTA) | ali.cheraghian@mq.edu.au | Homepage |